Nicholas A. Henry

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EDUCATION

Northeastern University — Boston, MA
Bachelor of Science in Mechanical Engineering and Computer Science — *May 2029*
Coursework**: Calculus III, Discrete Structures, Cornerstone of Engineering I and II* — *GPA: **3.94
***Activities**: Northeastern Electric Racing FSAE, Students for the Exploration and Discovery of Space*
Studied Abroad For Freshman Fall Semester At Queen's University Belfast

SKILLS

Technical Skills: Python, Flask, TensorFlow, Dash Plotly, ROS, Java, JavaScript, HTML/CSS, Linux, Docker, REST APIs, Git, Supabase, SQL, GitHub Actions, WordPress
Engineering Skills: Fusion360, SolidWorks, AutoCAD, MATLAB, Problem Solving, Cross-Functional Collaboration, 3D Printing, Laser Cutting, CNC Milling, MIG/TIG Welding
Certifications & Training: RECF Pre-Engineering Certification, Microsoft C# Certification

EXPERIENCE

Northeastern Electric Racing - *Python, Dash Plotly, Docker, Git, Figma* — Boston, MA
Lead SWE for Simulation Team — Jan 2026 – Present

- Lead on the simulation team developing a multi-body vehicle simulation tool for NER engineers to test and examine how changes to geometry and other variables go on to affect the speed and times of the car on custom tracks
- Use Python as the backend to do all the physics and Dash Plotly to display to a site to easily see all graphs and metrics worked on creating a large data structure of all the vehicle's variables to allow a tree visualization of dependency

Students for the Exploration and Discovery of Space - *Python, ROS 2, Git, Linux, Docker* — Boston, MA
SWE for Network Exploration Swarm Toolkit — Jan 2026 – Present

- Operate in NEST, a project to create a swarm of miniature C.O.B.R.A. (Crater Observing Bio-inspired Rolling Articulator) snake robots to explore and develop multi-agent collaboration in order to complete teamwork based tasks
- Developed the camera sensor ecosystem using a monocular simultaneous mapping program for multi-agent systems

SEONeeded - *Wordpress, SEOMoz, Semrush* — Miami, FL / Remote
Web Optimization Engineer — Oct 2025 – Present

- Perform client-facing tasks like identifying keywords, website maintenance, and tactics to target in order to drive more traffic for certain services increasing click-on rate and sales for 15 clients across the Greater South Florida region
- Adapted to the new wave of generative engine optimization space by learning what makes LLM models reference client's websites and how certain worded prompts get tailored responses, recommendations, and links to sources

Search Engine Optimization Intern — Jun 2025 – Aug 2025

- Drove a 54% increase in website traffic through targeted optimization efforts by leveraging WordPress and research

AI Camp - *HTML/CSS, JavaScript, Vite, React* — Remote
Web Development Intern — Sep 2023 – Nov 2023

- Defined, built, and optimized a new page of the website to allow for camp courses to be taught year-round, eliminating the seasonal limitations of the school year using HTML, CSS, JavaScript, and React
- Programmed with backend APIs gaining production level experience with authentication, scaling, and account management, facilitating a smooth user experience for 500+ users and counting

PROJECTS

Portfolio Website - *HTML/CSS, JavaScript, SQL, Supabase, GitHub Actions* — Present

- Built a site with HTML/CSS and Javascript hosted on Github Pages with a custom domain to show myself and projects
- Applied SQL databases with Supabase in order to have a global comment section where visitors can write something
- Use GitHub Actions in order to autoupdate the resume on the site to match this document as it changes in Google Docs

Text Transcriber - *Python, TensorFlow, Google APIs, Git* — Jul 2025

- Designed and trained an AI model with Python using TensorFlow on 100,000+ written character images to recognize and convert handwriting into text and to then convert into audio with Google TTS API as part of a 3-person team
- Tested with a group of visibly impaired people to see and hear writing and signage enabling them to be less dependant on caretakers in everyday life, with successful results in the limited cohort of 88% understanding the messages